

Evaluating the Transfer of RAG Poisoning Defenses to Multi-Hop Question Answering

Jeremiah O. Abimbola[†], Godlove Suila Kuaban*, Onyeka J. Nwobodo[†], Piotr Czekalski[†]

*Institute of Theoretical and Applied Informatics, Polish Academy of Sciences (IITIS-PAN),

Bałtycka 5, 44-100 Gliwice, Poland, Emails: gskuaban@iitis.pl

[†]Faculty of Automatic Control, Electronics and Computer Science, Silesian University of Technology,

Akademicka 16, 44-100 Gliwice, Poland, Email: {jeremiah.abimbola, onwobodo, piotr.czekalski}@polsl.pl

Abstract—Retrieval-Augmented Generation (RAG) systems improve the factuality of large language models by grounding responses in externally retrieved knowledge. However, this reliance on external corpora exposes RAG systems to knowledge-base poisoning attacks, in which adversaries inject malicious passages that influence retrieval and subsequent answer generation. While numerous defenses have been proposed to mitigate such attacks, existing evaluations have focused almost exclusively on single-hop question-answering benchmarks, where answers are typically supported by a single retrieved document. Consequently, it remains unclear whether these defenses generalize to multi-hop reasoning tasks that require aggregating evidence across multiple documents. This paper presents a systematic evaluation of defense transferability from single-hop to multi-hop RAG systems. We assess four representative defense mechanisms—FilterRAG, RAGDefender, RAGPart, and RAGMask—using four benchmark datasets spanning reasoning depths from one to four hops: Natural Questions, HotpotQA, 2WikiMultiHopQA, and MuSiQue. To stress-test defense behavior, we employ three established poisoning attacks together with a structured poisoning probe that mimics multi-step adversarial evidence chains. The evaluation comprises 240 experimental configurations and 48,000 query evaluations across multiple poisoning budgets. Our results show that reasoning depth alone provides little protection against poisoning attacks, with attack success rates remaining largely stable across single-hop and multi-hop settings. Defense effectiveness, however, varies substantially with reasoning complexity. FilterRAG consistently achieves the strongest protection, reducing attack success rates by up to 79.2% while maintaining acceptable answer quality. In contrast, defenses based on corpus partitioning, retrieval agreement, or token-level filtering exhibit reduced effectiveness in multi-hop environments, where answers depend on combining evidence across multiple retrieved passages. These findings demonstrate that robustness observed on single-hop benchmarks does not necessarily transfer to multi-hop reasoning tasks and highlight the need for security evaluations that account for realistic multi-document reasoning scenarios.

Index Terms—Retrieval-Augmented Generation (RAG), knowledge poisoning, multi-hop reasoning, RAG security, performance evaluations

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) has emerged as a widely adopted paradigm for enhancing the factuality and knowledge access capabilities of large language models (LLMs). By retrieving relevant documents from an external knowledge source and conditioning generation on the retrieved evidence, RAG systems can reduce hallucinations, improve answer accuracy, and incorporate information that may not be

present in the model’s parametric memory [1], [2]. As a result, RAG has become a foundational component in a growing range of applications, including question answering, enterprise search, decision support systems, and domain-specific assistants.

Despite these advantages, the reliance on external knowledge sources introduces a new attack surface. In particular, RAG systems are vulnerable to knowledge-base poisoning attacks, in which adversaries inject malicious or misleading passages into the retrieval corpus. Because retrieved documents directly influence the generation process, poisoned content can propagate through the RAG pipeline and alter the final output. Recent studies have demonstrated the effectiveness of such attacks under a variety of threat models and retrieval settings. PoisonedRAG [3] shows that adversarial passages can be optimized to maximize retrieval likelihood while remaining difficult to detect, whereas CorruptRAG [4] demonstrates that even a small number of carefully crafted injections can substantially influence model behavior. Similarly, PIDP [5] combines corpus poisoning with prompt-based manipulation to further amplify adversarial influence. Collectively, these findings indicate that retrieval-based grounding alone is insufficient to guarantee robustness against adversarially manipulated knowledge sources.

To address this challenge, a growing body of work has proposed defenses that operate at different stages of the RAG pipeline. Representative examples include FilterRAG [6], which filters suspicious passages using statistical quality signals; RAGDefender [7], which verifies retrieved evidence through cross-retrieval consistency; RAGPart [8], which reduces exposure to poisoned content through corpus partitioning; and RAGMask [9], which applies token-level filtering during generation. While these methods have demonstrated promising results, their evaluation has been conducted predominantly in single-hop question-answering settings, where answers are often supported by a single retrieved document.

This evaluation setting leaves an important gap in our understanding of RAG security. Many real-world information-seeking tasks require *multi-hop reasoning*, in which the answer must be derived by retrieving, combining, and reasoning over evidence distributed across multiple documents. Benchmarks such as HotpotQA [10], 2WikiMultiHopQA [11], and MuSiQue [12] exemplify this setting by requiring models to

construct intermediate reasoning chains before arriving at a final answer. In contrast to single-hop retrieval, multi-hop reasoning introduces complex dependencies among retrieved passages, creating new opportunities for adversarial influence and potentially invalidating assumptions made by existing defense mechanisms. For example, a defense that removes suspicious passages in isolation may inadvertently eliminate critical supporting evidence, while a poisoned document may influence the reasoning process indirectly through intermediate inference steps. Consequently, it remains unclear whether defenses developed and validated in single-hop settings retain their effectiveness when applied to multi-hop retrieval and reasoning tasks.

This paper investigates the transferability of existing RAG defenses across reasoning depths. Rather than proposing a new attack or defense mechanism, we perform a systematic evaluation of established defenses under both single-hop and multi-hop question-answering conditions. We employ representative poisoning probes to stress-test defense behavior across four benchmark datasets spanning reasoning depths from one to four hops: Natural Questions (NQ), HotpotQA, 2WikiMultiHopQA, and MuSiQue. The study evaluates four defense mechanisms, four poisoning probes, and three poisoning budgets, resulting in 240 experimental configurations and 48,000 query evaluations.

The goal is to answer a fundamental question: *Do defenses developed for single-hop RAG systems transfer effectively to multi-hop reasoning environments?* Through extensive experiments, we show that defense effectiveness varies substantially across reasoning depths. Our findings highlight the importance of evaluating RAG security under realistic single-hop and multi-document reasoning conditions.

The main contributions of this paper are as follows:

- We present the first systematic study of defense transferability from single-hop to multi-hop RAG systems under corpus-poisoning attacks.
- We evaluate four representative RAG defense mechanisms across four benchmark datasets spanning reasoning depths from one to four hops using a comprehensive experimental grid of 240 configurations and 48,000 query evaluations.
- We analyze the interaction between reasoning depth, poisoning strategy, and defense mechanism, identifying conditions under which existing defenses succeed or fail.
- We demonstrate that defense effectiveness is strongly influenced by reasoning structure, providing evidence that evaluation solely on single-hop benchmarks may overestimate real-world robustness.

II. RELATED WORK

A. RAG Poisoning Attacks

Retrieval-Augmented Generation (RAG) systems are susceptible to corpus poisoning attacks that target the external knowledge base used during retrieval. In these attacks, adversaries inject carefully crafted passages into the retrieval

corpus with the objective of maximizing their likelihood of being retrieved for specific queries and subsequently influencing the generated response. PoisonedRAG [3] formulates this threat as an optimization problem, generating poisoned passages that simultaneously maximize retrieval relevance and minimize detectability through measures such as fluency and perplexity. Experimental results demonstrate attack success rates exceeding 90% on benchmarks such as Natural Questions in single-hop question-answering settings.

Building on this line of research, CorruptRAG [4] investigates more constrained and stealthy poisoning strategies. Rather than relying on a large number of adversarial passages, it shows that a small set of carefully designed injections can effectively manipulate model outputs when the corresponding queries trigger retrieval of the poisoned content. Similarly, PIDP [5] combines corpus poisoning with prompt-based manipulation, demonstrating that adversarial influence can be distributed across both the user query and the retrieved context. Collectively, these studies highlight that retrieval-based grounding alone does not provide inherent robustness against adversarial manipulation and that RAG systems remain vulnerable to carefully engineered poisoning attacks.

Despite these advances, existing evaluations predominantly focus on single-hop question answering, where the final response is often determined by information contained within a single retrieved passage. Consequently, the effectiveness of corpus poisoning attacks in multi-hop reasoning scenarios remains insufficiently understood. In multi-hop question answering, models must retrieve, integrate, and reason across multiple documents to derive the correct answer, potentially altering both the attack surface and the effectiveness of existing poisoning strategies. This limitation motivates the present study, which investigates whether corpus-poisoning attacks and corresponding defense mechanisms developed for single-hop RAG systems transfer effectively to multi-hop question-answering tasks that require evidence aggregation across multiple retrieved documents.

B. RAG Defense Mechanisms

A variety of defense mechanisms have been proposed to mitigate the impact of poisoned or misleading content in Retrieval-Augmented Generation (RAG) systems. These defenses generally operate at either the retrieval stage or the generation stage of the RAG pipeline. At the retrieval stage, FilterRAG [6] employs filtering techniques based on passage-quality indicators, including perplexity and semantic consistency measures, to identify and remove potentially malicious documents before they are supplied to the language model. RAGPart [8] adopts a structural defense strategy by partitioning the retrieval corpus into smaller subsets, thereby reducing the likelihood that poisoned content disproportionately influences retrieval outcomes across a wide range of queries.

At the generation stage, RAGDefender [7] introduces a cross-verification framework that compares evidence obtained from multiple retrieval sources to detect inconsistencies and

identify potentially compromised passages. In contrast, RAG-Mask [9] enhances robustness during answer generation by selectively masking or down-weighting tokens that exhibit characteristics associated with adversarial manipulation. Together, these approaches represent complementary strategies for improving the resilience of RAG systems against corpus-poisoning attacks.

C. Multi-Hop Question Answering

Multi-hop question answering (QA) requires models to retrieve, integrate, and reason over information distributed across multiple sources in order to generate a correct answer. Unlike single-hop QA, where a single supporting passage is often sufficient, multi-hop reasoning involves a sequence of intermediate inference steps that connect multiple pieces of evidence. Consequently, the correctness of the final answer depends not only on the relevance of individual retrieved passages but also on the model’s ability to aggregate and reason across them effectively.

Several benchmark datasets have been developed to evaluate multi-hop reasoning capabilities. HotpotQA [10] and 2WikiMultiHopQA [11] focus primarily on two-hop reasoning, where answering a question requires linking two related facts originating from different documents. MuSiQue [12] further increases reasoning complexity by constructing compositional questions that require multiple sequential reasoning steps, often involving up to four evidence hops. These benchmarks have played a central role in advancing retrieval and reasoning techniques for complex question-answering tasks.

III. METHODOLOGY

A. Threat Model

We adopt a corpus-poisoning threat model consistent with prior studies on RAG security [3], [4]. Let $\mathcal{C}$ denote the original knowledge corpus, $\mathcal{Q}$ the set of user queries, and $R(\cdot)$ and $G(\cdot)$ the retriever and generator components of the RAG system, respectively. The adversary is assumed to have write access to the retrieval corpus and can inject a bounded set of poisoned passages, denoted by $\mathcal{P}_{adv}$, into the corpus. Following existing work, we consider attack budgets of $|\mathcal{P}_{adv}| \in \{1, 3, 5\}$, representing low-, medium-, and high-capacity poisoning scenarios.

At inference time, given a user query $q \in \mathcal{Q}$, the retriever operates over the poisoned corpus and returns the top- k retrieved passages:

$$\mathcal{D}_q = R(q, \mathcal{C} \cup \mathcal{P}_{adv}). \quad (1)$$

The generator then produces an answer conditioned on both the query and the retrieved context:

$$\hat{y} = G(q, \mathcal{D}_q). \quad (2)$$

The attacker’s objective is to manipulate the retrieval and generation process such that the model produces a predetermined target answer y_{adv} or otherwise follows an attacker-

specified behavior. An attack is considered successful when the generated output satisfies the adversarial objective, formally:

$$\hat{y} = y_{adv}. \quad (3)$$

We assume a black-box attack setting in which the adversary cannot modify the parameters of either the retriever or the language model and has no direct control over the inference procedure beyond the injected corpus content. Consequently, the attack is realized solely through manipulation of the external knowledge source. Consistent with prior defense evaluations, mitigation mechanisms are applied either at the retrieval stage or at the generation stage, while the underlying retriever and language model remain unchanged.

B. Datasets

We evaluate the proposed framework on four widely used question-answering benchmarks that span a range of reasoning complexities, from single-hop retrieval to multi-hop compositional reasoning. As summarized in Table I, these datasets enable a systematic analysis of how poisoning attacks and defense mechanisms behave as the number of required reasoning steps increases.

Natural Questions (NQ) [13] serves as the single-hop baseline, where answers can typically be derived from a single relevant passage. In contrast, HotpotQA [10] and 2WikiMultiHopQA [11] require two-hop reasoning, necessitating the retrieval and integration of evidence from multiple supporting documents. MuSiQue [12] further increases reasoning complexity by constructing compositional questions that require up to four sequential reasoning hops. Together, these benchmarks provide a diverse evaluation environment for examining the robustness of RAG systems under increasing reasoning depth and evidence aggregation requirements.

TABLE I
EVALUATION DATASETS SPANNING DIFFERENT REASONING DEPTHS.

Dataset	Reasoning Hops	Corpus Size	Queries
Natural Questions (NQ)	1	2.68M	200
HotpotQA	2	5.23M	200
2WikiMultiHopQA	2	430K	200
MuSiQue	4	139K	200

The selected datasets allow us to investigate whether the effectiveness of existing RAG defenses degrades as reasoning complexity increases. In particular, they provide a controlled setting for evaluating how poisoning attacks propagate through multi-document reasoning chains and whether defense mechanisms developed for single-hop retrieval scenarios generalize to multi-hop question-answering tasks.

C. RAG Pipeline

We adopt a standard retrieve-and-generate architecture consisting of a dense retriever and a frozen large language model (LLM) generator. For document retrieval, we employ `Contriever` [14], a dense bi-encoder retriever that maps queries and passages into a shared embedding space. Passage

embeddings are indexed using FAISS [15] to enable efficient approximate nearest-neighbor search over the corpus.

Given an input query q , the retriever returns the top- k most relevant passages, where $k = 5$. The retrieved passages are concatenated and provided as contextual evidence to the generator. For answer generation, we use Mistral-7B-Instruct-v0.3 [16], which remains frozen throughout all experiments. To reduce output variability and ensure reproducibility, decoding is performed with a temperature of $T = 0.1$ and a maximum generation length of 256 tokens.

This configuration is kept fixed across all datasets, attack settings, and defense mechanisms to ensure that observed performance differences are attributable to the evaluated attacks and defenses rather than changes in the underlying RAG architecture.

D. Poisoning Probes

The poisoning methods considered in this study are used as evaluation probes to assess the robustness and transferability of existing RAG defense mechanisms across different reasoning depths. Our objective is not to introduce new attack techniques, but rather to examine whether defenses that are effective in single-hop retrieval settings maintain their effectiveness in multi-hop question-answering scenarios. To this end, we employ three representative poisoning attacks from the literature together with a structured poisoning variant designed to emulate reasoning-oriented adversarial evidence.

- **PoisonedRAG** [3]: An optimization-based corpus-poisoning attack that generates adversarial passages to maximize retrieval likelihood while preserving linguistic fluency and maintaining low detectability.
- **CorruptRAG** [4]: A stealth-oriented poisoning strategy that relies on a small number of carefully crafted adversarial passages designed to exert maximal influence while minimizing corpus perturbation.
- **PIDP** [5]: A hybrid attack that combines corpus poisoning with prompt-based manipulation, enabling adversarial influence to be distributed across both the retrieved context and the user query.
- **Structured Poisoning Probe (SPP)**: A structured poisoning variant that distributes misleading information across multiple interconnected factual statements rather than embedding misinformation within a single isolated passage. The probe is designed to resemble multi-step evidence chains commonly encountered in multi-hop reasoning tasks. It is included solely as an evaluation instrument to investigate whether existing defenses exhibit different behavior when adversarial content mimics the structure of legitimate reasoning paths.

Together, these probes capture a diverse set of poisoning characteristics, including retrieval optimization, stealth, prompt-context interaction, and reasoning-structured misinformation. This diversity enables a comprehensive assessment of whether defense effectiveness changes as reasoning complexity increases.

E. Defense Mechanisms

We evaluate four representative defense mechanisms that operate at different stages of the Retrieval-Augmented Generation (RAG) pipeline. Collectively, these defenses cover retrieval-time filtering, corpus-level isolation, cross-source verification, and generation-time token filtering. Let $\mathcal{D}_q$ denote the set of passages retrieved for a query q . A defense mechanism is represented by a transformation function $\phi(\cdot)$ that modifies either the retrieved context, the retrieval process itself, or intermediate representations prior to answer generation.

- **FilterRAG** [6]: A retrieval-stage defense that assigns each retrieved passage $d \in \mathcal{D}_q$ a quality score $s(d, q)$ based on indicators such as perplexity and semantic consistency. Passages whose scores fall below a predefined threshold τ are removed before generation:

$$\phi(\mathcal{D}_q) = \{d \in \mathcal{D}_q : s(d, q) \geq \tau\}.$$

- **RAGDefender** [7]: A retrieval-verification defense that evaluates the agreement between dense and sparse retrieval signals. Passages exhibiting substantial disagreement between retrieval modalities are treated as potentially suspicious and excluded from the final context supplied to the generator.
- **RAGPart** [8]: A corpus-partitioning defense that decomposes the knowledge base into disjoint subsets,

$$\mathcal{C} = \bigcup_i \mathcal{C}_i,$$

and restricts retrieval to a query-dependent partition selected through a routing mechanism. By limiting the retrieval search space, the defense reduces the likelihood that poisoned documents influence unrelated queries.

- **RAGMask** [9]: A generation-stage defense that applies token-level filtering to the retrieved context. Tokens identified as anomalous according to a reference language model are masked or down-weighted before answer generation, thereby reducing the influence of potentially adversarial content.

These defenses were selected because they represent distinct design philosophies and intervention points within the RAG pipeline, enabling a comprehensive evaluation of defense transferability across varying reasoning depths.

F. Experimental Setup

To systematically evaluate defense transferability, we consider all combinations of datasets, poisoning probes, defense mechanisms, and poisoning budgets. The resulting experimental grid is defined as

$$\mathcal{E} = \mathcal{D}_{\text{sets}} \times \mathcal{A} \times \mathcal{F} \times \mathcal{B}, \quad (4)$$

where $\mathcal{D}_{\text{sets}}$ denotes the set of evaluation datasets, $\mathcal{A}$ the set of poisoning probes, $\mathcal{F}$ the set of defense mechanisms, and $\mathcal{B} = \{1, 3, 5\}$ the poisoning budgets corresponding to the number of injected adversarial passages.

This design yields a total of 240 experimental configurations, comprising 48 no-defense baseline configurations and 192 active-defense configurations. Each configuration is evaluated on 200 queries sampled from the corresponding benchmark dataset, resulting in 48,000 query evaluations overall.

The no-defense configurations establish baseline attack success rates (ASRs) for each dataset, poisoning method, and attack budget. These baseline measurements are reported directly in the ASR comparison tables and serve as the reference point for quantifying the effectiveness of each defense. For active-defense configurations, we report both the resulting ASR and the corresponding reduction relative to the no-defense baseline, thereby enabling a consistent comparison of defense performance across different reasoning depths and attack settings.

G. Evaluation Metrics

To assess both security robustness and utility preservation, we employ a set of complementary evaluation metrics that capture attack effectiveness, defense performance, retrieval behavior, and answer quality.

1) *Attack Success Rate (ASR)*: Attack Success Rate (ASR) measures the proportion of queries for which the attack successfully induces the target adversarial output. Formally,

$$\text{ASR} = \mathbb{E}_{q \sim \mathcal{Q}} [\mathbb{I}(\hat{y}_q = y_q^{adv})], \quad (5)$$

where $\hat{y}_q$ is the generated answer for query q , y_q^{adv} is the attacker-specified target output, and $\mathbb{I}(\cdot)$ denotes the indicator function. Higher ASR values indicate more successful attacks and, consequently, lower system robustness.

2) *ASR Reduction*: To quantify the effectiveness of a defense mechanism, we measure the relative reduction in ASR compared with the corresponding no-defense baseline:

$$\text{ASR}_{\text{red}} = \frac{\text{ASR}_{\text{base}} - \text{ASR}_{\text{def}}}{\text{ASR}_{\text{base}} + \epsilon}, \quad (6)$$

where ASR_{base} and ASR_{def} denote the attack success rates without and with defense, respectively, and ϵ is a small constant introduced to avoid numerical instability when ASR_{base} approaches zero. Larger values of ASR_{red} indicate more effective defenses.

3) *Retrieval Success Rate (RSR)*: Retrieval Success Rate (RSR) measures the frequency with which at least one poisoned passage appears among the retrieved documents. This metric captures the exposure of adversarial content at the retrieval stage:

$$\text{RSR} = \mathbb{E}_{q \sim \mathcal{Q}} [\mathbb{I}(\mathcal{D}_q \cap \mathcal{P}_{adv} \neq \emptyset)], \quad (7)$$

where $\mathcal{D}_q$ denotes the set of retrieved passages for query q and $\mathcal{P}_{adv}$ represents the injected poisoned passages. Lower RSR values indicate that poisoned content is less likely to reach the generator.

4) *Answer Quality (F1)*: To evaluate utility preservation, we compute the token-level F1 score between generated and reference answers on non-target queries. The F1 score balances precision and recall and is widely used in open-domain question-answering benchmarks. Higher F1 values indicate better answer quality and greater preservation of the system’s original functionality under defense.

5) *Stealthiness*: Stealthiness measures the ability of poisoned passages to evade detection by a defense mechanism. It is defined as the fraction of adversarial passages that remain unflagged:

$$\text{Stealthiness} = 1 - \frac{|\{d \in \mathcal{P}_{adv} : \phi(d) = 1\}|}{|\mathcal{P}_{adv}|}, \quad (8)$$

where $\phi(d) = 1$ indicates that passage d has been identified as suspicious by the defense. A higher stealthiness value indicates that adversarial content is more difficult to detect and remove.

Collectively, these metrics provide a comprehensive evaluation of both attack and defense behavior. ASR and RSR quantify attack effectiveness and adversarial exposure, ASR Reduction measures defense efficacy, F1 assesses utility preservation, and Stealthiness characterizes the detectability of poisoned content.

TABLE II
DEFENSE EFFECTIVENESS (ASR REDUCTION %) ACROSS REASONING DEPTHS. HIGHER IS BETTER. NEGATIVE VALUES INDICATE THAT THE DEFENSE INCREASES ASR RELATIVE TO THE NO-DEFENSE BASELINE.

Defense	NQ (1-hop)	HotpotQA (2-hop)	2WikiMQA (2-hop)	MuSiQue (4-hop)
FilterRAG	38.0%	79.2%	75.2%	58.1%
RAGDefender	9.3%	34.4%	22.2%	15.4%
RAGPart	-0.9%	-0.5%	-1.3%	-0.5%
RAGMask	12.1%	17.3%	2.6%	10.2%
Average	14.6%	32.6%	24.7%	20.8%

TABLE III
AVERAGE ASR BY REASONING DEPTH (NO DEFENSE). MULTI-HOP REASONING DOES NOT SUBSTANTIALLY REDUCE ATTACK SUCCESS IN THIS EVALUATION.

Attack	1-hop	2-hop	4-hop	Δ (4-1)
PoisonedRAG	0.537	0.595	0.538	+0.002
CorruptRAG	0.318	0.084	0.237	-0.082
PIDP	0.880	0.897	0.797	-0.083
SPP	0.545	0.709	0.612	+0.067
Average	0.570	0.571	0.546	-0.024

IV. RESULTS

A. Defense Transfer Across Reasoning Depths

Table II summarizes defense effectiveness, measured as Attack Success Rate (ASR) reduction, across datasets requiring increasing levels of reasoning complexity. Overall, we observe substantial variation in defense transferability. While some

TABLE IV
ASR UNDER EACH DEFENSE-PROBE COMBINATION (AVERAGED ACROSS ALL DATASETS AND INJECTION LEVELS).

Defense	Poisoned	Corrupt	PIDP	SPP
No Defense	0.566	0.181	0.868	0.644
FilterRAG	0.141	0.066	0.360	0.274
RAGDefender	0.240	0.148	0.767	0.642
RAGPart	0.570	0.184	0.871	0.652
RAGMask	0.494	0.178	0.690	0.656

TABLE V
SECURITY-UTILITY TRADE-OFF. IDEAL DEFENSES APPEAR IN THE TOP ROW (HIGH ASR REDUCTION) WITH RIGHTMOST COLUMNS (LOW F1 DEGRADATION).

Defense	ASR Reduction	F1 Score	F1 Change
No Defense	—	0.328	—
FilterRAG	62.7%	0.298	-9.1%
RAGDefender	20.3%	0.310	-5.5%
RAGPart	-0.8%	0.315	-3.8%
RAGMask	10.6%	0.345	+5.2%

defenses retain effectiveness as reasoning depth increases, others exhibit limited or even negative impact relative to the no-defense baseline.

FilterRAG consistently provides the strongest protection across all datasets, achieving ASR reductions ranging from 38.0% to 79.2%. Notably, its effectiveness remains high in multi-hop settings, suggesting that filtering based on passage-level quality signals generalizes well beyond the single-hop conditions under which many defenses are typically evaluated.

RAGDefender provides moderate protection, with ASR reductions ranging from 9.3% to 34.4%. Although beneficial in most cases, its effectiveness varies considerably across datasets, indicating sensitivity to the structure of retrieved evidence and the characteristics of the poisoning attack.

In contrast, RAGPart consistently exhibits negative ASR reduction, implying that attack success rates are slightly higher than those observed without defense. This behavior is particularly noteworthy in multi-hop settings, where successful reasoning often depends on retrieving evidence distributed across multiple regions of the corpus. Restricting retrieval through corpus partitioning may therefore disrupt legitimate evidence aggregation while failing to prevent retrieval of locally relevant poisoned content.

RAGMask achieves only modest improvements, with reductions between 2.6% and 17.3%. While token-level filtering can suppress localized adversarial signals, its effectiveness appears limited when misinformation is incorporated into a broader reasoning chain.

Taken together, these results indicate that defense transferability is highly dependent on the underlying defense mechanism. Methods operating on passage-level quality signals transfer more successfully to multi-hop reasoning environments than approaches based on corpus partitioning or token-level filtering.

B. Attack Success Across Reasoning Depths

Table III reports attack success rates in the absence of defenses. Contrary to the intuition that deeper reasoning may inherently increase robustness, we observe only minor changes in ASR as reasoning depth increases.

Across all poisoning probes, the average ASR decreases from 0.570 in the single-hop setting to 0.546 in the four-hop setting, corresponding to a reduction of only 2.4 percentage points. Although individual attacks exhibit some variability, the overall trend remains largely flat. PoisonedRAG and SPP maintain comparable effectiveness across reasoning depths, while CorruptRAG and PIDP show only moderate decreases.

These findings suggest that increasing reasoning complexity alone does not provide meaningful protection against corpus-poisoning attacks. The ability of poisoned content to influence retrieval and generation appears largely preserved even when answers require multiple intermediate reasoning steps.

C. Defense–Probe Interaction Analysis

To better understand the robustness of individual defenses, Table IV reports attack success rates for each defense–probe combination averaged across all datasets and injection budgets.

FilterRAG is the only defense that consistently reduces ASR across all four poisoning probes. Its effectiveness extends not only to optimization-based attacks such as PoisonedRAG, but also to structured attacks such as SPP. This consistency suggests that passage-level filtering captures adversarial characteristics that remain detectable regardless of attack strategy.

RAGDefender achieves moderate reductions against PoisonedRAG and CorruptRAG but provides limited protection against PIDP and SPP. In particular, SPP produces structured passages that appear highly relevant under both dense and sparse retrieval, thereby weakening the effectiveness of agreement-based verification.

RAGPart and RAGMask exhibit minimal improvements across most probes. Their performance indicates that restricting retrieval scope or filtering individual tokens is insufficient when adversarial influence is embedded within the broader retrieval and reasoning process.

Overall, the results highlight that defense effectiveness depends not only on reasoning depth but also on the structural properties of the poisoning attack itself.

D. Security–Utility Trade-Off

A practical defense must reduce attack success while preserving answer quality. Table V summarizes this trade-off using ASR reduction and F1 performance on benign queries.

FilterRAG achieves the strongest overall security benefit, reducing ASR by 62.7% while incurring a relatively modest F1 degradation of 9.1%. Although some utility loss is observed, the magnitude of the security improvement suggests a favorable trade-off.

RAGDefender provides moderate security gains with smaller utility degradation, whereas RAGMask slightly improves F1 performance by removing noisy tokens but delivers only limited security benefits. RAGPart performs poorly on

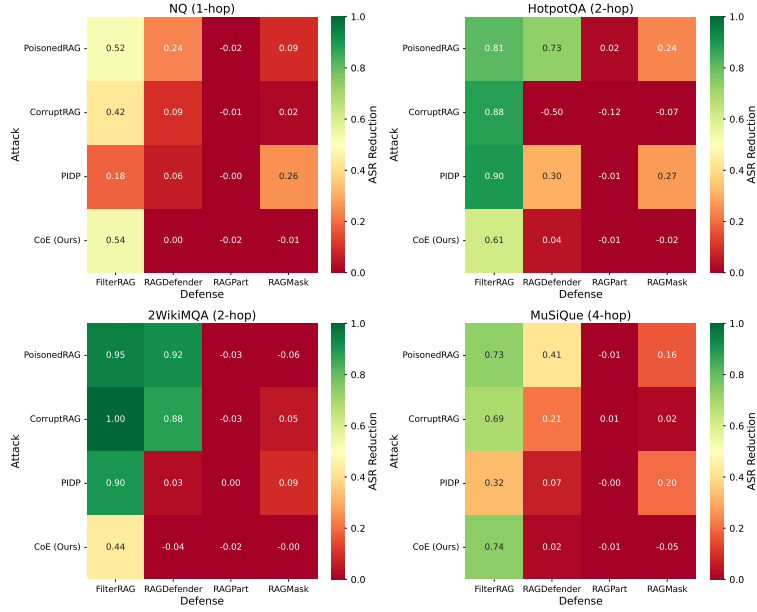


Fig. 1. Defense effectiveness by dataset and attack. FilterRAG consistently achieves the largest ASR reductions, whereas RAGPart frequently produces negative reductions relative to the no-defense baseline.

both dimensions, reducing neither attack effectiveness nor utility degradation.

These findings indicate that effective defense transfer requires balancing robustness against adversarial content with preservation of legitimate evidence needed for reasoning. Among the evaluated approaches, FilterRAG provides the most favorable balance between security and utility.

E. Why Defenses Fail in Multi-Hop Settings

Our results suggest that many defense failures stem from assumptions that hold in single-hop retrieval settings but become less reliable when reasoning requires aggregating evidence across multiple documents.

One prominent example is corpus isolation. RAGPart assumes that restricting retrieval to a query-specific partition reduces exposure to poisoned content. However, multi-hop reasoning frequently requires evidence distributed across different regions of the corpus. Limiting retrieval in this manner can therefore disrupt valid reasoning chains while leaving relevant poisoned passages accessible within the selected partition.

A second limitation arises from token-centric defenses such as RAGMask. These approaches assume that adversarial behavior manifests through suspicious lexical patterns. In multi-hop settings, however, failures often emerge at the reasoning level rather than the token level. Once an incorrect premise enters the reasoning chain, subsequent inference steps may propagate and reinforce the error even when individual tokens appear benign.

Agreement-based defenses such as RAGDefender face a related challenge. Their effectiveness depends on the assumption that adversarial passages produce inconsistent retrieval signals. Structured poisoning attacks, however, can generate coherent

multi-step narratives that appear relevant under multiple retrieval modalities simultaneously. In such cases, agreement among retrieval signals no longer serves as a reliable indicator of trustworthiness.

In contrast, FilterRAG appears less sensitive to reasoning depth because it operates on statistical properties of retrieved passages rather than on assumptions regarding evidence structure. Features such as unusual entity co-occurrence patterns, synthetic discourse structures, and distributional irregularities remain detectable even when adversarial content is embedded within multi-hop reasoning chains.

F. Injection Budget Analysis

Increasing the poisoning budget from one to five injected passages consistently improves attack effectiveness across all poisoning probes. On average, ASR increases by approximately 15 percentage points as the number of injected passages grows.

Nevertheless, even the smallest attack budget remains highly effective. Depending on the poisoning strategy, a single injected passage achieves attack success rates between 40% and 70%, demonstrating that corpus poisoning remains practical under limited adversarial resources.

Interestingly, defense effectiveness remains relatively stable across injection budgets. Although larger budgets increase attack success overall, they do not substantially alter the relative ranking of defenses. This observation suggests that defense transferability is driven primarily by reasoning structure and attack characteristics rather than by the absolute number of injected passages.

V. CONCLUSION

This paper investigated the transferability of existing RAG defenses from single-hop to multi-hop question-answering environments under corpus-poisoning attacks. Although a growing body of research has proposed defenses against knowledge-base poisoning, their evaluation has been conducted predominantly on single-hop benchmarks, leaving open the question of whether such defenses remain effective when answers require reasoning across multiple retrieved documents.

To address this gap, we performed a comprehensive empirical study involving four benchmark datasets spanning reasoning depths from one to four hops, four representative defense mechanisms, four poisoning probes, and multiple poisoning budgets. The resulting evaluation comprised 240 experimental configurations and 48,000 query evaluations. Our findings show that increasing reasoning depth alone does not substantially reduce attack success rates, indicating that multi-hop reasoning should not be regarded as an inherent defense against retrieval poisoning. At the same time, defense effectiveness varies considerably across reasoning settings. Among the evaluated methods, FilterRAG consistently provided the strongest protection and demonstrated the best transferability to multi-hop environments. In contrast, defenses based on corpus partitioning, retrieval agreement, or token-level filtering exhibited limited robustness when answers depended on aggregating evidence across multiple documents.

The results suggest that many existing defenses implicitly rely on assumptions that hold in single-hop retrieval but become less reliable in multi-hop reasoning scenarios. In particular, defenses that evaluate passages independently may struggle when adversarial influence is distributed across evidence chains or when legitimate reasoning requires integrating information from multiple sources. These findings highlight the importance of evaluating RAG security under realistic multi-document reasoning conditions and suggest that robustness claims based solely on single-hop benchmarks may overestimate practical security.

This work has several limitations. Our evaluation focused on a specific set of retrieval architectures, language models, poisoning attacks, and defense mechanisms. Future studies should examine whether the observed trends generalize to larger language models, alternative retrievers, adaptive attackers, and more diverse reasoning tasks. In addition, the structured poisoning probe considered in this work represents only one form of reasoning-oriented adversarial content. Developing stronger multi-hop poisoning attacks and corresponding defense strategies remains an important direction for future research.

Overall, our results demonstrate that defense transferability cannot be assumed across reasoning depths. As RAG systems are increasingly deployed in applications requiring complex multi-document reasoning, security evaluations must move beyond single-hop benchmarks and account for the unique challenges introduced by multi-hop retrieval and evidence

aggregation.

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